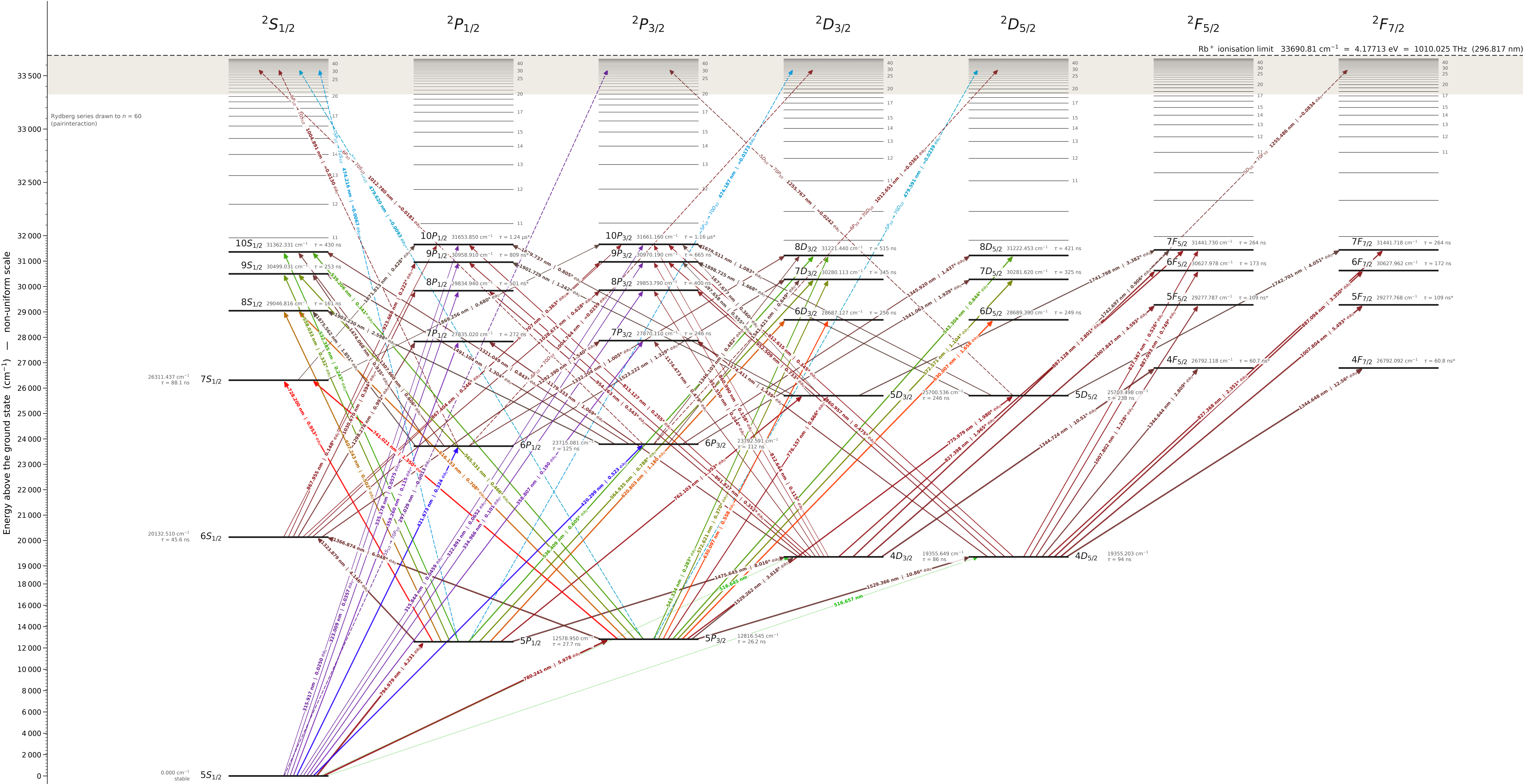
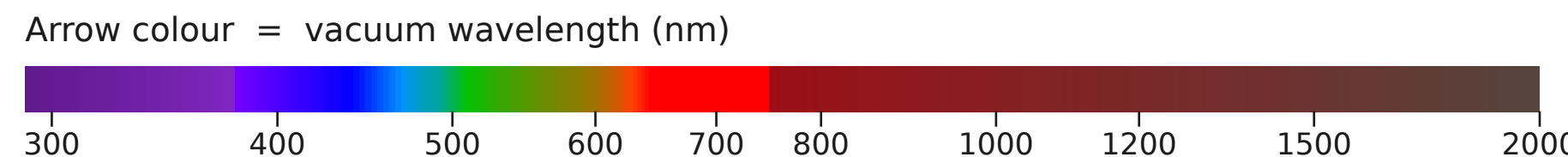


# Rubidium-85

<sup>85</sup>Rb (Rb I), nuclear spin  $I = 5/2$ . Energy levels and transitions below 2  $\mu\text{m}$ : 36 levels, 94 lines.

Arrow label:  $\langle U||e\mathbf{r}||J' \rangle$  in ea<sub>0</sub>  
Matrix elements are measured or NIST values; \* = high-accuracy theory,  $\approx$  = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: NIST energy and lifetime (measured where a value exists, otherwise ARC, marked  $\approx$ ).



## Key transitions

| Transition                            | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $\tilde{\nu}$ (cm <sup>-1</sup> ) | $\langle U  e\mathbf{r}  J' \rangle$ (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | $\Gamma/2\pi$ up. (MHz) | branch (%) | Typical use                                            |
|---------------------------------------|--------------------|--------------------|-------------|-----------------------------------|---------------------------------------------------------|----------------------|-------------------------|------------|--------------------------------------------------------|
| 5S <sub>1/2</sub> - 5P <sub>3/2</sub> | 780.2414           | 780.0267           | 384.2304    | 12816.547                         | 5.978                                                   | 3.811e+07            | 6.065                   | 100.0      | D2: MOT, imaging, Rydberg lower leg                    |
| 5S <sub>1/2</sub> - 5P <sub>1/2</sub> | 794.9790           | 794.7604           | 377.1074    | 12578.948                         | 4.231                                                   | 3.610e+07            | 5.746                   | 100.0      | D1: optical pumping, $\Lambda$ / gray-molasses cooling |
| 5S <sub>1/2</sub> - 6P <sub>3/2</sub> | 420.2989           | 420.1805           | 713.2839    | 23792.591                         | 0.523                                                   | 1.866e+06            | 1.421                   | 21.1       | blue leg of 420 + 1013 Rydberg excitation              |
| 5S <sub>1/2</sub> - 6P <sub>1/2</sub> | 421.6726           | 421.5539           | 710.9602    | 23715.081                         | 0.324                                                   | 1.414e+06            | 1.273                   | 17.4       | blue leg (421 + 1005)                                  |
| 5P <sub>3/2</sub> - 5D <sub>3/2</sub> | 775.9786           | 775.7651           | 386.3411    | 12886.953                         | 1.980*                                                  | 2.833e+06            | 0.667                   | 66.8       | 780 + 776 ladder; 5S-5D two-photon at 778.1 nm         |
| 5P <sub>1/2</sub> - 5D <sub>3/2</sub> | 762.1030           | 761.8932           | 393.3752    | 13121.586                         | 1.353*                                                  | 2.095e+06            | 0.646                   | 51.0       | 795 + 762 ladder                                       |
| 5P <sub>3/2</sub> - 4D <sub>3/2</sub> | 1529.3658          | 1528.9480          | 196.0240    | 6538.658                          | 10.86*                                                  | 1.114e+07            | 1.693                   | 100.0      | telecom C-band ladder                                  |
| 5P <sub>3/2</sub> - 4D <sub>5/2</sub> | 1529.2615          | 1528.8437          | 196.0374    | 6539.104                          | 3.618*                                                  | 1.854e+06            | 1.851                   | 15.5       | telecom C-band ladder                                  |
| 5P <sub>1/2</sub> - 4D <sub>3/2</sub> | 1475.6447          | 1475.2415          | 203.1603    | 6776.699                          | 8.016*                                                  | 1.013e+07            | 1.851                   | 84.5       | telecom S-band ladder                                  |
| 5P <sub>3/2</sub> - 6S <sub>1/2</sub> | 1366.8737          | 1366.5000          | 219.3271    | 7315.965                          | 6.048*                                                  | 1.451e+07            | 3.493                   | 65.9       | telecom E-band ladder                                  |
| 5P <sub>1/2</sub> - 6S <sub>1/2</sub> | 1323.8791          | 1323.5171          | 226.4500    | 7553.560                          | 4.146*                                                  | 7.506e+06            | 3.493                   | 34.1       | telecom O-band ladder                                  |
| 5P <sub>3/2</sub> - 6D <sub>3/2</sub> | 630.0068           | 629.8326           | 475.8559    | 15872.845                         | 1.658                                                   | 3.712e+06            | 0.639                   | 77.9       | red ladder                                             |
| 5P <sub>3/2</sub> - 7S <sub>1/2</sub> | 741.0211           | 740.8171           | 404.5667    | 13494.892                         | 1.350*                                                  | 4.536e+06            | 1.807                   | 40.0       | ladder to 7S                                           |
| 6S <sub>1/2</sub> - 7P <sub>3/2</sub> | 1292.3904          | 1292.0370          | 231.9674    | 7737.600                          | 1.540*                                                  | 5.565e+05            | 0.647                   | 14.2       | O-band, from 6S                                        |
| 4D <sub>5/2</sub> - 4F <sub>7/2</sub> | 1344.6483          | 1344.2807          | 222.9523    | 7436.889                          | 12.56*                                                  | 1.643e+07            | 2.618                   | 98.2       | strongest line of the diagram                          |
| 6P <sub>3/2</sub> - 7D <sub>5/2</sub> | 1541.0626          | 1540.6416          | 194.5362    | 6489.029                          | 1.929*                                                  | 3.433e+05            | 0.490                   | 11.5       | telecom C-band, from 6P                                |
| 6P <sub>3/2</sub> - 8S <sub>1/2</sub> | 1903.2303          | 1902.7107          | 157.5177    | 5254.225                          | 2.598*                                                  | 9.918e+05            | 0.989                   | 16.0       | longest 6P line below 2 $\mu\text{m}$                  |

$\lambda$  and  $\nu$  from NIST level energies (centre of gravity, no hyperfine structure);  $\lambda$  air uses standard air (Ciddor).  
 $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ . Branch: share of the upper level's decays that takes this line.

## Rydberg excitation wavelengths (vacuum, nm)

| Lower $\rightarrow$ series                        | n = 30   | n = 40   | n = 50   | n = 70   | n = 100  | n $\rightarrow$ $\infty$ | d <sub>no</sub> pairint. | d <sub>no</sub> ARC |
|---------------------------------------------------|----------|----------|----------|----------|----------|--------------------------|--------------------------|---------------------|
| 5S <sub>1/2</sub> $\rightarrow$ nP <sub>3/2</sub> | 298.113  | 297.510  | 297.247  | 297.029  | 296.918  | 296.817                  | 0.0013                   | 0.0023              |
| 5P <sub>1/2</sub> $\rightarrow$ nS <sub>1/2</sub> | 477.099  | 475.483  | 474.788  | 474.216  | 473.927  | 473.667                  | 0.0067                   | 0.0067              |
| 5P <sub>1/2</sub> $\rightarrow$ nD <sub>3/2</sub> | 476.682  | 475.318  | 474.707  | 474.187  | 473.917  | 473.667                  | 0.0173                   | 0.0166              |
| 5P <sub>3/2</sub> $\rightarrow$ nS <sub>1/2</sub> | 482.570  | 480.916  | 480.205  | 479.620  | 479.324  | 479.059                  | 0.0093                   | 0.0095              |
| 5P <sub>3/2</sub> $\rightarrow$ nD <sub>3/2</sub> | 482.143  | 480.747  | 480.122  | 479.591  | 479.315  | 479.059                  | 0.0239                   | 0.0230              |
| 4D <sub>5/2</sub> $\rightarrow$ nP <sub>3/2</sub> | 704.765  | 701.404  | 699.946  | 698.736  | 698.121  | 697.564                  | 0.0100                   | 0.0092              |
| 4D <sub>5/2</sub> $\rightarrow$ nF <sub>7/2</sub> | 703.548  | 700.913  | 699.701  | 698.649  | 698.092  | 697.564                  | 0.0559                   | 0.0582              |
| 6S <sub>1/2</sub> $\rightarrow$ nP <sub>3/2</sub> | 745.611  | 741.850  | 740.219  | 738.866  | 738.178  | 737.556                  | 0.0055                   | 0.0061              |
| 6P <sub>1/2</sub> $\rightarrow$ nS <sub>1/2</sub> | 1017.930 | 1010.598 | 1007.464 | 1004.891 | 1003.596 | 1002.433                 | 0.0130                   | 0.0130              |
| 6P <sub>1/2</sub> $\rightarrow$ nD <sub>3/2</sub> | 1016.033 | 1009.855 | 1007.100 | 1004.764 | 1003.554 | 1002.433                 | 0.0258                   | 0.0253              |
| 6P <sub>3/2</sub> $\rightarrow$ nS <sub>1/2</sub> | 1026.025 | 1018.576 | 1015.393 | 1012.780 | 1011.464 | 1010.283                 | 0.0181                   | 0.0181              |
| 6P <sub>3/2</sub> $\rightarrow$ nD <sub>3/2</sub> | 1024.097 | 1017.821 | 1015.023 | 1012.651 | 1011.421 | 1010.283                 | 0.0361                   | 0.0354              |
| 5D <sub>5/2</sub> $\rightarrow$ nP <sub>3/2</sub> | 1275.375 | 1264.411 | 1259.681 | 1255.767 | 1253.782 | 1251.986                 | 0.0242                   | 0.0236              |
| 5D <sub>5/2</sub> $\rightarrow$ nF <sub>7/2</sub> | 1271.394 | 1262.817 | 1258.887 | 1255.486 | 1253.687 | 1251.986                 | 0.0834                   | 0.0846              |
| 7S <sub>1/2</sub> $\rightarrow$ nP <sub>3/2</sub> | 1382.573 | 1369.698 | 1364.148 | 1359.560 | 1357.233 | 1355.129                 | 0.0114                   | 0.0118              |
| 4F <sub>7/2</sub> $\rightarrow$ nD <sub>5/2</sub> | 1478.152 | 1465.114 | 1459.322 | 1454.424 | 1451.889 | 1449.545                 | 0.0186                   | 0.0186              |
| 7P <sub>3/2</sub> $\rightarrow$ nS <sub>1/2</sub> | 1764.032 | 1742.128 | 1732.838 | 1725.240 | 1721.425 | 1718.006                 | 0.0291                   | 0.0291              |
| 7P <sub>3/2</sub> $\rightarrow$ nD <sub>5/2</sub> | 1758.339 | 1739.920 | 1731.758 | 1724.865 | 1721.300 | 1718.006                 | 0.0491                   | 0.0485              |

Rydberg term energies from pairinteraction, lower level from NIST. d<sub>no</sub> =  $\langle U||e\mathbf{r}||J' \rangle$  to n = 70 in ea<sub>0</sub> (calculated).  
S  $\rightarrow$  nP values come from a near-cancelling integral: treat them as an order of magnitude.

## <sup>85</sup>Rb hyperfine structure ( $I = 5/2$ )

| Level             | A (MHz)     | B (MHz)    | Shift of each F level from the centre of gravity (MHz)                 |
|-------------------|-------------|------------|------------------------------------------------------------------------|
| 5S <sub>1/2</sub> | 1011.910815 | -          | F=2: -1770.84 F=3: +1264.89                                            |
| 5P <sub>1/2</sub> | 120.605(29) | -          | F=2: -211.06 F=3: +150.76                                              |
| 5P <sub>3/2</sub> | 25.0401(11) | 26.000(22) | F=1: -113.26 F=2: -83.98 F=3: -20.56 F=4: +100.40                      |
| 4D <sub>5/2</sub> | -4.983(7)   | 6.70(21)   | F=0: +48.29 F=1: +41.70 F=2: +29.12 F=3: +11.76 F=4: -8.57 F=5: -29.47 |
| 4D <sub>3/2</sub> | 7.33(4)     | 4.52(23)   | F=1: -35.31 F=2: -24.27 F=3: -4.32 F=4: +28.61                         |
| 6S <sub>1/2</sub> | 239.069(26) | -          | F=2: -418.37 F=3: +298.84                                              |
| 6P <sub>1/2</sub> | 39.122(9)   | -          | F=2: -68.46 F=3: +48.90                                                |
| 6P <sub>3/2</sub> | 8.171(7)    | 8.16(4)    | F=1: -37.19 F=2: -27.37 F=3: -6.53 F=4: +32.68                         |
| 5D <sub>3/2</sub> | 4.26990(20) | 1.9106(8)  | F=1: -21.08 F=2: -14.07 F=3: -2.12 F=4: +16.49                         |
| 5D <sub>5/2</sub> | -2.2112(12) | 2.680(20)  | F=0: +21.22 F=1: +18.37 F=2: +12.90 F=3: +5.30 F=4: -3.70 F=5: -13.15  |
| 7S <sub>1/2</sub> | 94.6813(15) | -          | F=2: -165.69 F=3: +118.35                                              |
| 7P <sub>1/2</sub> | 17.68(8)    | -          | F=2: -30.94 F=3: +22.10                                                |
| 7P <sub>3/2</sub> | 3.710(10)   | 3.68(8)    | F=1: -16.90 F=2: -12.43 F=3: -2.95 F=4: +14.83                         |
| 6D <sub>3/2</sub> | 2.30(4)     | 1.62(6)    | F=1: -10.94 F=2: -7.64 F=3: -1.47 F=4: +9.03                           |
| 6D <sub>5/2</sub> | -1.069(18)  | -0.4(4)    | F=0: +9.07 F=1: +8.10 F=2: +6.12 F=3: +3.06 F=4: -1.19 F=5: -6.78      |
| 7D <sub>3/2</sub> | 1.350(20)   | 0.31(6)    | F=1: -6.87 F=2: -4.42 F=3: -0.51 F=4: +5.14                            |
| 7D <sub>5/2</sub> | -0.55(10)   | -          | F=0: +4.81 F=1: +4.26 F=2: +3.16 F=3: +1.51 F=4: -0.69 F=5: -3.44      |

A, B: magnetic-dipole and electric-quadrupole constants (measured; see data file for sources).